THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2241 Introduction to Mathematical Analysis
Test 1
9 October 2020

1. (7 points) Let $S = \left\{ \dfrac{2ab}{a^2+b^2} : a, b \in \mathbb{N} \right\}$. Prove that $\sup S = 1$ and $\inf S = 0$. (You may use any method introduced in Section 1.)

2. (7 points) Let $\{a_n\}$ be a convergent sequence of real numbers. Let $\{b_n\}$ be a sequence of real numbers such that
$$|a_n - b_n| < \frac{1}{n}$$
for any $n \in \mathbb{N}$. Using the Cauchy criterion, prove that $\{b_n\}$ converges. (If you prove this result using other methods, you may get some partial points.)

3. (7 points) Let $c \geqslant 0$ be a real number. Define the sequence $\{a_n\}$ of real numbers as follows:
$$a_1 = c, \quad a_{n+1} = \frac{3a_n^2}{a_n^2+2} \text{ for any } n \in \mathbb{N}.$$
For each value of c, find the limit of $\{a_n\}$ or show that the sequence diverges.

4. (7 points) Define the sequence $\{b_n\}$ of real numbers as follows:
$$b_1 = 1, \quad b_{n+1} = \frac{b_n}{2} + \frac{1}{b_n^7} \text{ for any } n \in \mathbb{N}.$$
Prove that $\limsup_{n\to\infty} b_n$ and $\liminf_{n\to\infty} b_n$ exist as real numbers.

5. Prove or disprove each of the following. State clearly whether it is true or false.

   (a) (4 points) Let $\{a_n\}$ be a sequence of real numbers. If each of the subsequences $\{a_{3n}\}$, $\{a_{3n-1}\}$ and $\{a_{3n-2}\}$ converges, then $\{a_n\}$ must be convergent.

   (b) (4 points) Let $\{b_n\}$ be a convergent sequence of real numbers. Then the sequence $\{c_n\}$ defined by $c_n = b_n - b_{n^2}$ for any $n \in \mathbb{N}$ converges to 0.

   (c) (4 points) Let X be a nonempty set of nonzero real numbers such that $\sup X$ exists and $\sup X \neq 0$. Define
   $$Y = \left\{ \frac{1}{x} : x \in X \right\}.$$
   Then $\inf Y$ must exist and $\inf Y = \dfrac{1}{\sup X}$.